

COE305 MACHINE LEARNING PROJECT GUIDELINES

Project Weight: 30 % of Final Grade

Team Size: 3 students

1. Project Objective

The purpose of this project is to apply the concepts learned throughout the course to a **real-world regression or classification problem**.

You will work as a team to design, implement, and evaluate several machine-learning models from the course topics.

By completing this project, you will:

- Gain experience with the full ML pipeline — data → model → evaluation → interpretation.
- Apply supervised and ensemble methods in practical scenarios.
- Develop teamwork, communication, and report-writing skills.

2. Timeline & Deliverables

Stage	Description	Due Week	Weight
Stage 1 – Project Proposal	Submit a 1-page proposal including team details, problem statement, dataset source, and objectives.	Week 3	5%
Stage 2 – Dataset & EDA Submission	Submit cleaned dataset and EDA report (visualizations + summary stats).	Week 5	5 %
Stage 3 – Midterm Progress Report	Submit initial models (≥ 3 algorithms) + baseline results.	Week 7	10 %
Stage 4 – Final Report & Presentation	Submit final report and 10-minute presentation slides, 1-page summary of personal contribution and learning.	Week 14	10% Includes 2% for Individual Reflection

3. Project Instructions

A. Problem Definition

- Choose a **supervised learning problem** (regression or classification).
- Clearly state the goal and why it matters.
- Make sure the problem is feasible with your resources.

B. Dataset

You may use:

1. Public datasets (Kaggle, UCI, OpenML, Google Dataset Search)
2. Combined datasets or API-based data collection
3. Self-collected datasets (eligible for bonus points)

Your report must include:

- Dataset source and link
- Number of instances and features
- Target variable and problem type
- Ethical and privacy considerations

C. Data Pre-processing & EDA

- Handle missing values, duplicates, and outliers.
- Encode categorical variables and scale numerical ones.
- Provide visual summaries (correlation plots, histograms, box plots).

D. Model Implementation

- Train at least **three models** from different families (e.g., Decision Tree, Random Forest, SVM).
- Use **cross-validation** and **hyperparameter tuning** (GridSearchCV / RandomizedSearchCV).
- Evaluate with appropriate metrics (Accuracy, Precision, Recall, F1, RMSE, R^2 , etc.).

E. Results and Discussion

- Compare models using tables and graphs.
- Explain why certain models performed better.
- Discuss overfitting/underfitting and future improvements.

F. Report Format

Your final report (PDF) must include:

1. Title of Project

2. Team Members and IDs
3. Introduction & Motivation
4. Dataset Description
5. Methodology (Preprocessing + Algorithms)
6. Results & Discussion
7. Conclusion & Future Work
8. References

G. Presentation

- Each team gives a **10-minute presentation** in Week 14.
- All members must participate.
- Visuals should clearly show key findings and metrics.

4. Bonus Opportunities

- +3 % for self-collected or combined datasets.
- +2 % for creative visualizations or simple UI (e.g., Streamlit dashboard).